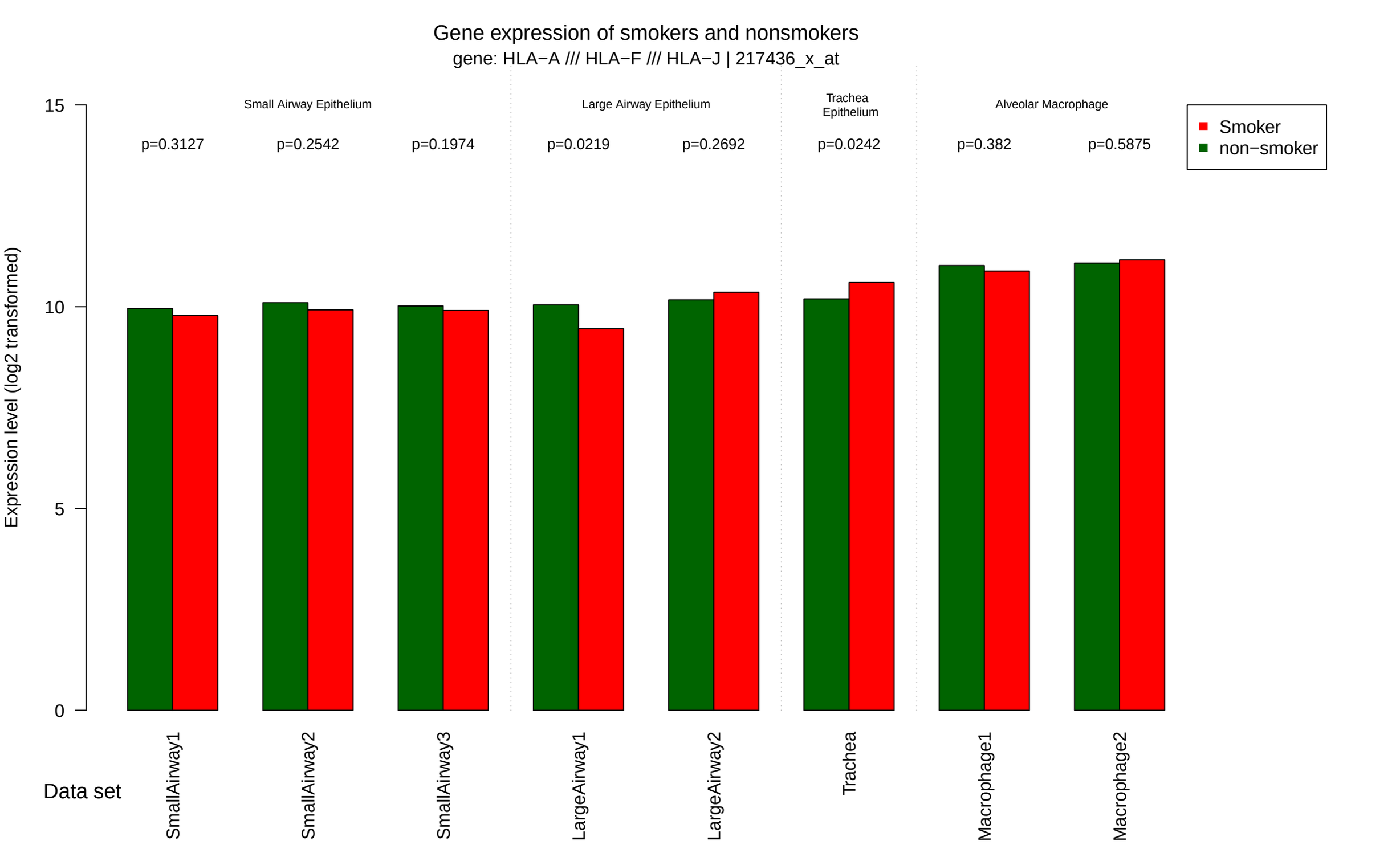
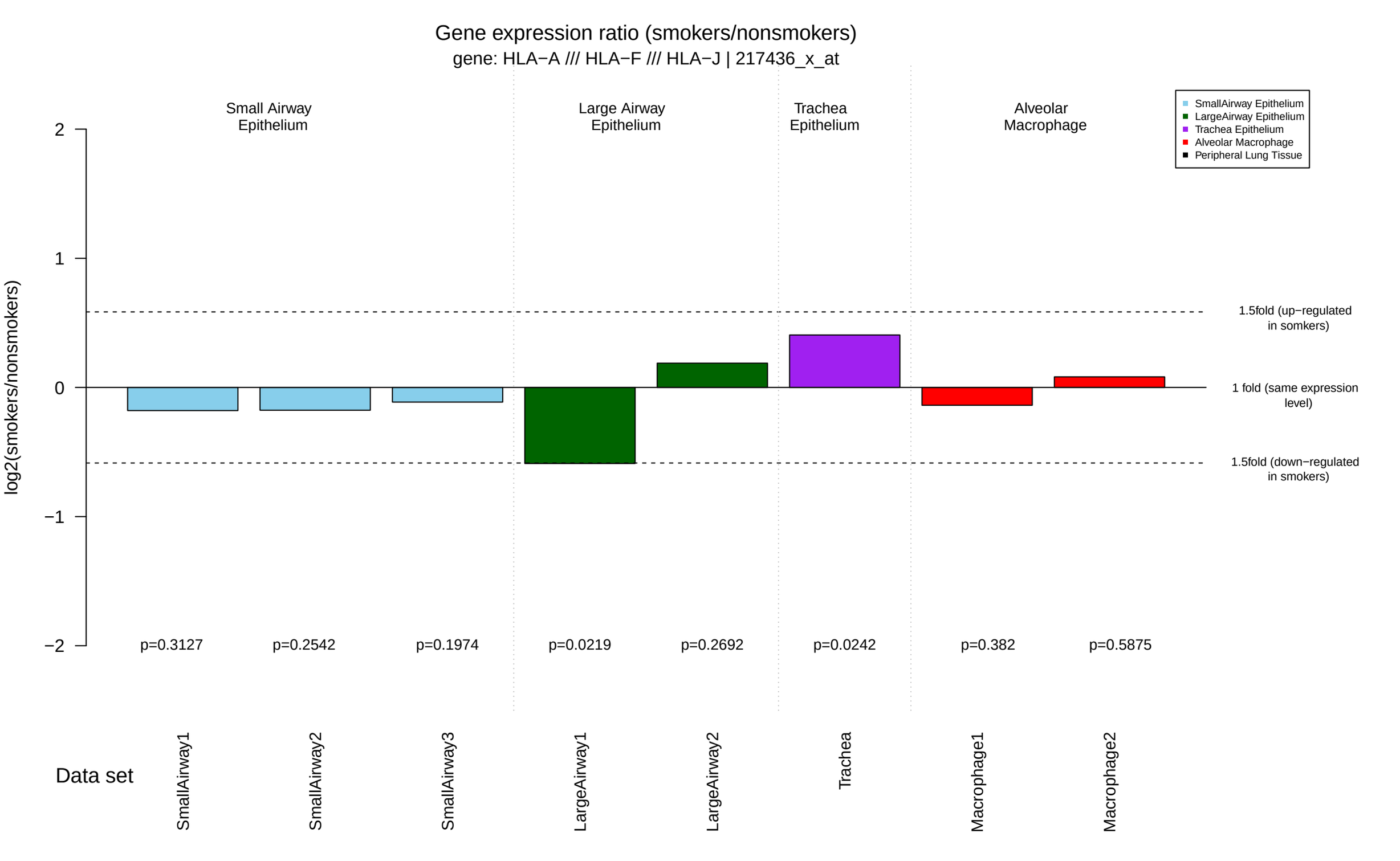
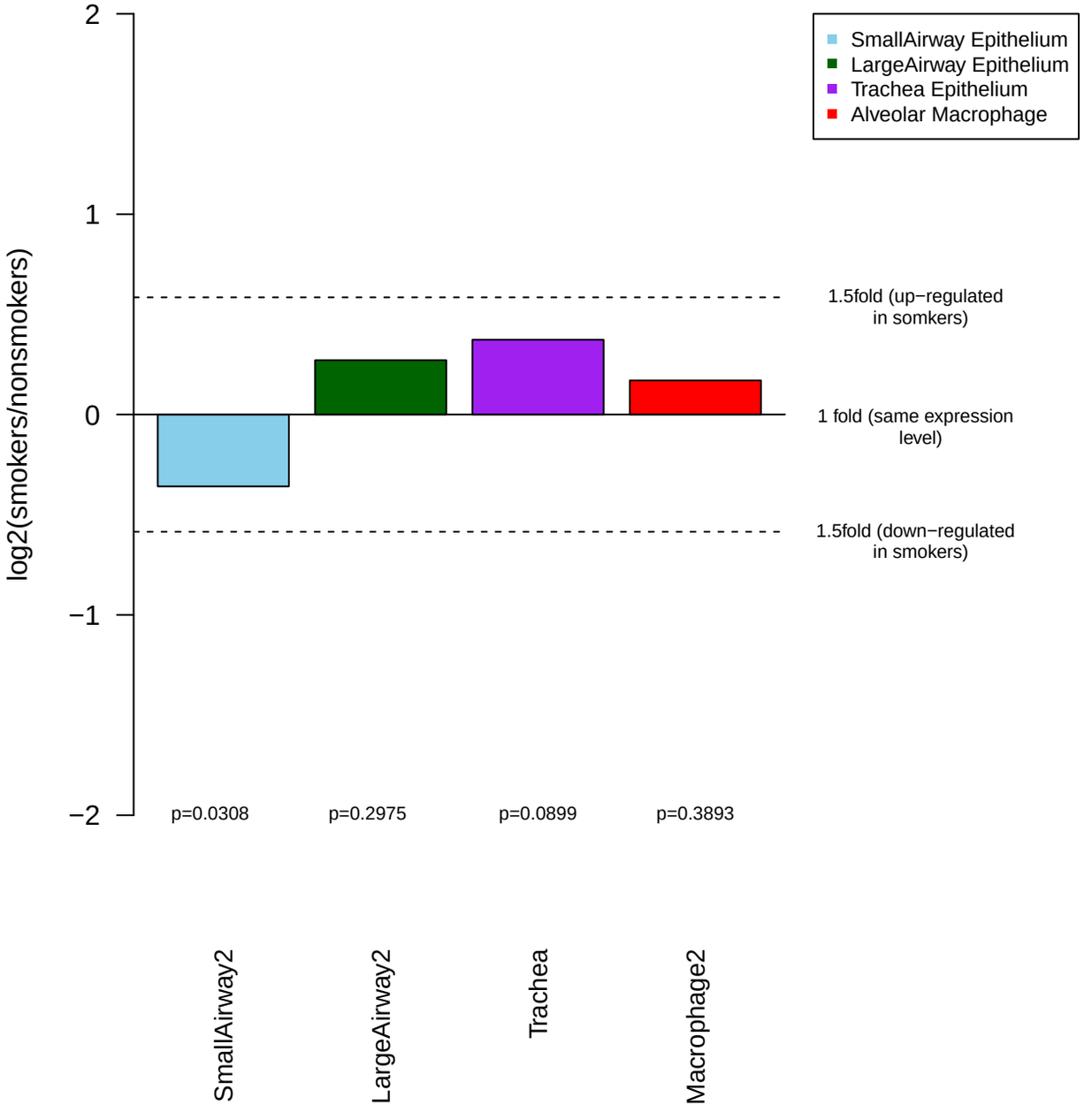

Gene expression of smokers and nonsmokers
gene: HLA−A /// HLA−F /// HLA−J | 217436_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3127
p=0.2542
p=0.1974
p=0.0219
p=0.2692
p=0.0242
p=0.382
p=0.5875
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: HLA−A /// HLA−F /// HLA−J | 217436_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3127
p=0.2542
p=0.1974
p=0.0219
p=0.2692
p=0.0242
p=0.382
p=0.5875
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

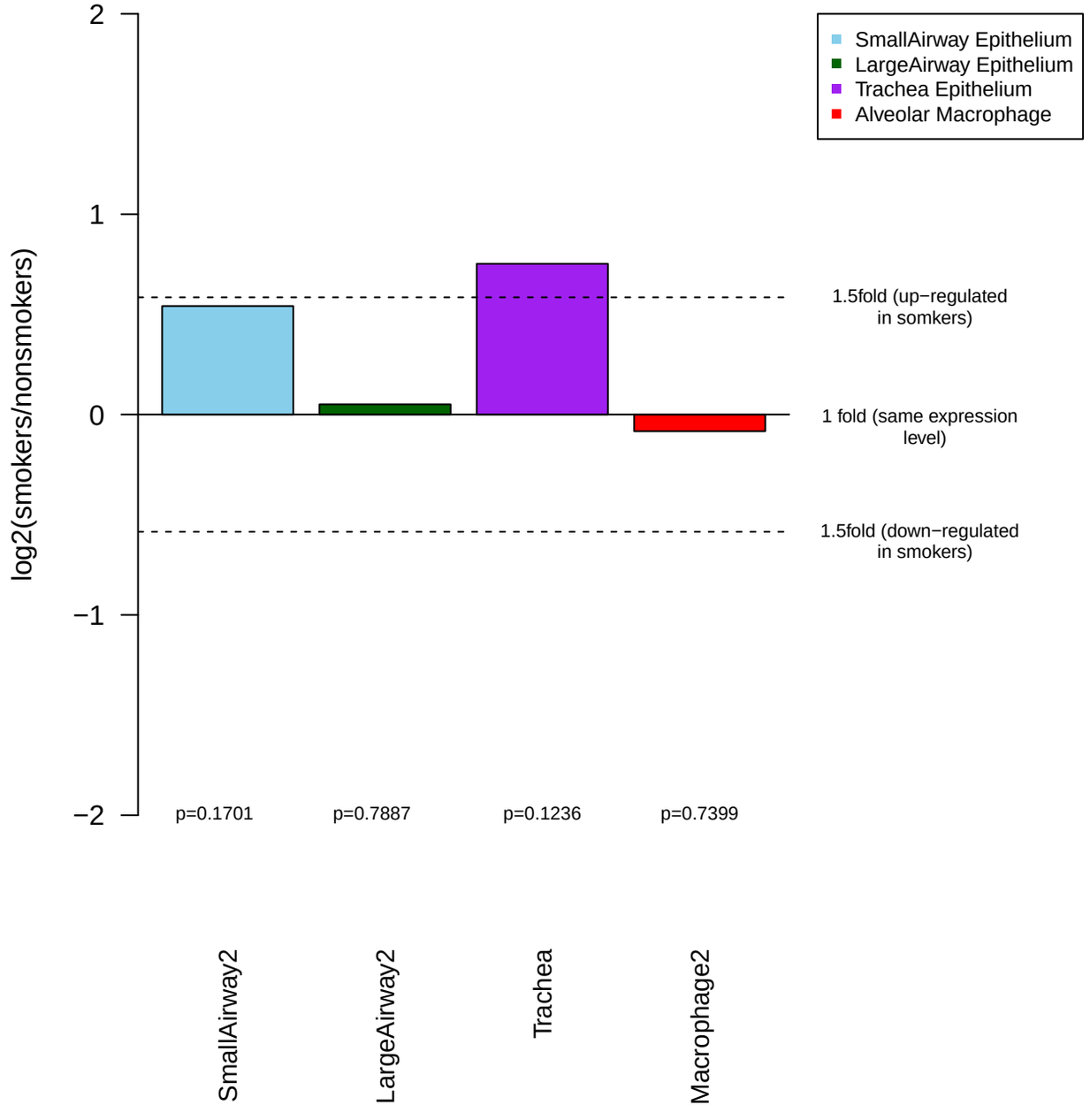
Gene expression ratio in Males

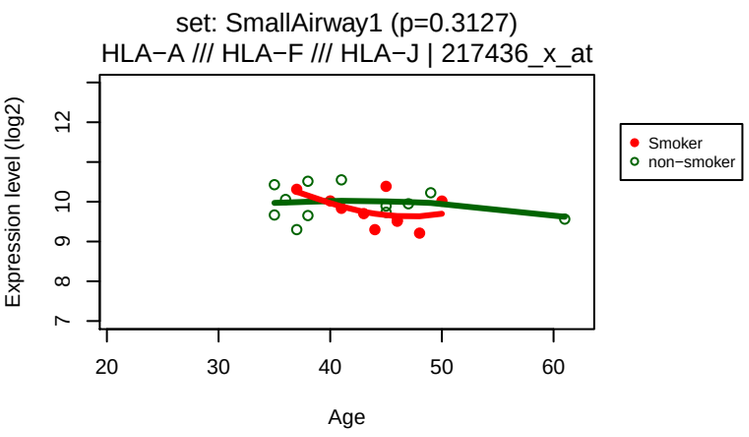
gene: HLA−A /// HLA−F /// HLA−J | 217436_x_at

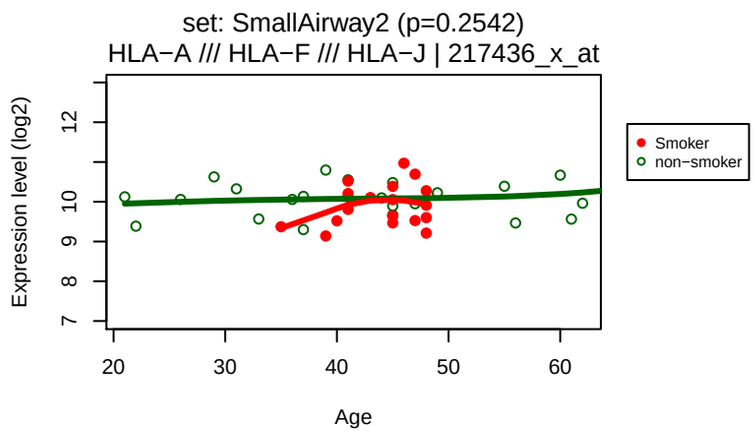


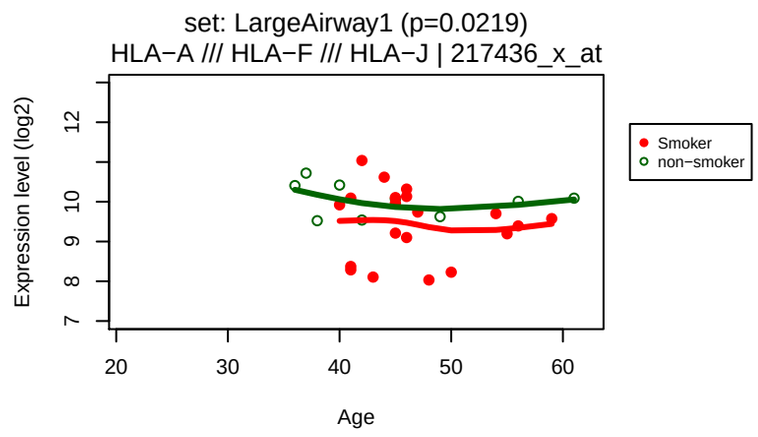
Gene expression ratio in Females

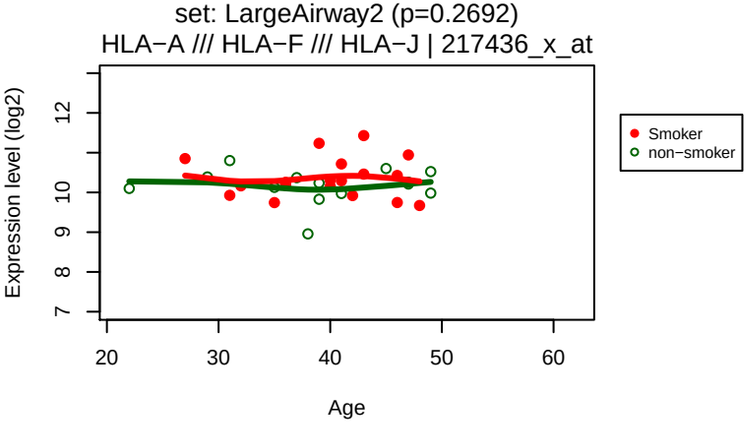
gene: HLA−A /// HLA−F /// HLA−J | 217436_x_at

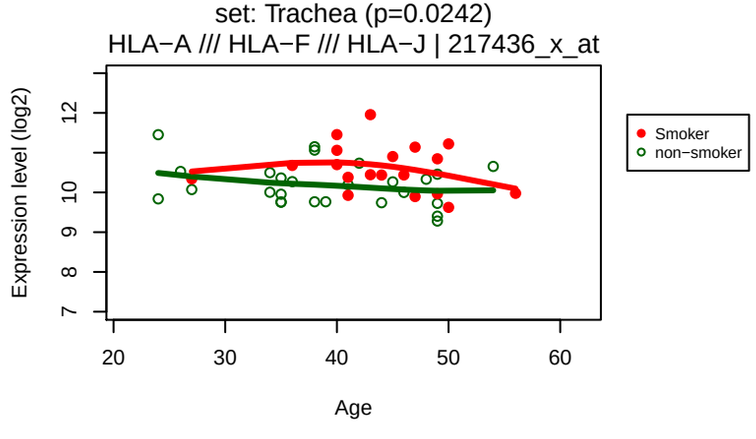


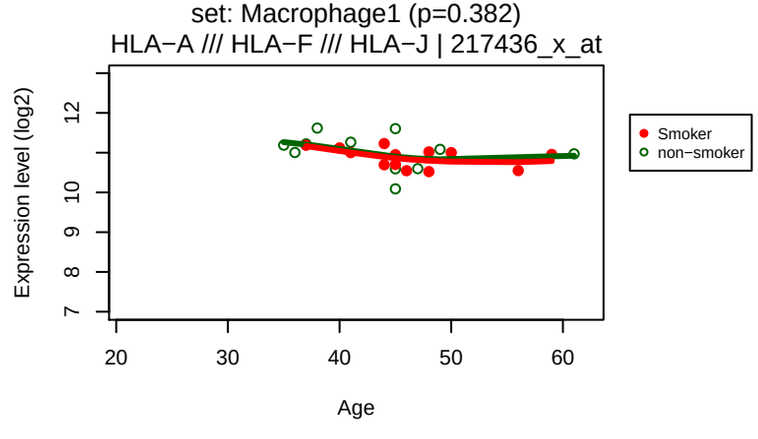

set: SmallAirway1 (p=0.3127)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker

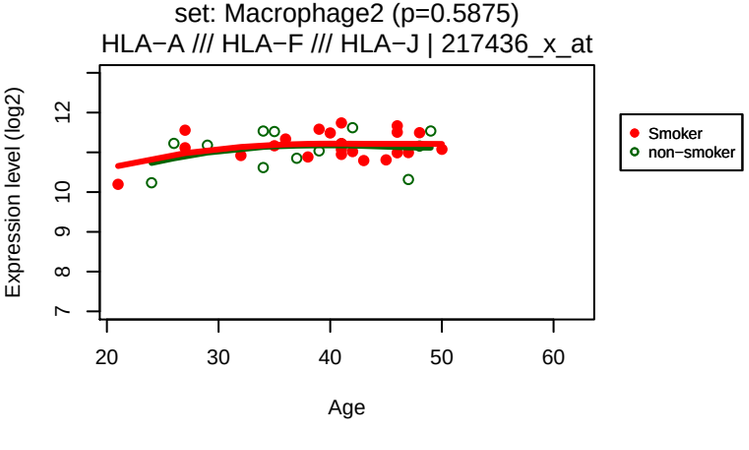

set: SmallAirway2 (p=0.2542)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker

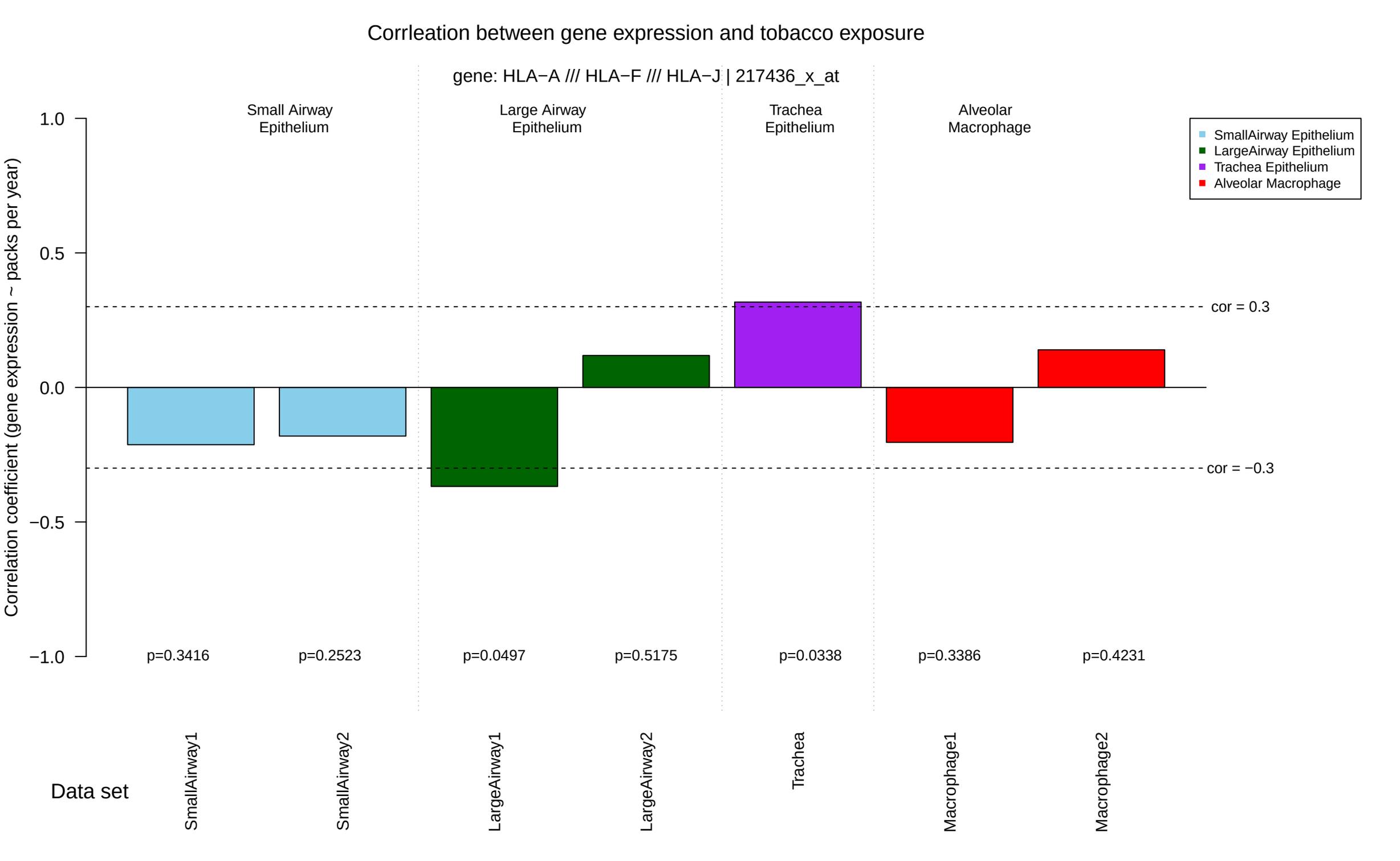

set: LargeAirway1 (p=0.0219)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2692)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0242)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.382)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5875)
HLA−A /// HLA−F /// HLA−J | 217436_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−A /// HLA−F /// HLA−J | 217436_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3416
p=0.2523
p=0.0497
p=0.5175
p=0.0338
p=0.3386
p=0.4231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2